IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 3 June 2025, accepted 27 June 2025, date of publication 2 July 2025, date of current version 10 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3585180

RESEARCH ARTICLE

TriNet: A Hybrid Feature Integration Approach for Motor Imagery Classification in Brain–Computer Interface

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This work was supported in part by the Education and Teaching Reform Research Project of Northwestern Polytechnical University under Grant 2024JGZ19.

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ABSTRACT Brain-computer interface (BCI) is inevitably a promising technology holding the potential to revolutionize the world with its wide range of applications. From healthcare to innovative computer gaming, integrating BCI for intelligent control has become an emergent scope. However, optimizing motor imagery (MI) classification in non-invasive BCI remains a significant challenge due to the poor quality of the acquired signal. In this paper, we propose a unified approach for MI classification by combining features from three diverse domains. Initially, the EEG data is preprocessed using bandpass filtering to extract the relevant EEG signals. Next, the preprocessed signal is fed simultaneously to three branches to extract three distinct categories of features from the signal. Specifically, spectral features are extracted using the fast-Fourier transform (FFT) and a spatial transformer is utilized to extract spatial features from the EEG data. Moreover, the third branch extracts temporal features using an encoder-decoder architecture. The features obtained using the three branches are concatenated together to obtain a comprehensive features set which is finally classified using extreme learning machine (ELM). Our proposed approach which uses a novel combination of features from three distinct domains is hereby named TriNet and is validated using two benchmark datasets BCI Competition IV-2a and BCI Competition IV-2b. The experimental results show an accuracy of 87.30% and 92.64% respectively on BCI IV-2a and BCI IV-2b datasets in subject-specific classification. Moreover, TriNet is also tested in subject-independent classification setup, and average classification accuracies of 63.92% and 78.60% are obtained on BCI IV-2a and 2b datasets respectively which is an improvement of 8 to 10% compared to the existing methods. The classification performance and computational cost comparisons demonstrate the superior performance of TriNet compared to the existing methods highlighting its potential to enhance MI classification in BCI.

INDEX TERMS TriNet, brain–computer interface (BCI), electroencephalography (EEG), motor imagery (MI), fast Fourier transform (FFT), self-attention, extreme learning machine (ELM), encoder-decoder.

I. INTRODUCTION

The past few decades have witnessed an unprecedented growth in the healthcare industry in terms of data availability, accuracy, and smart healthcare services [1], [2]. Electroencephalography (EEG) is one of the most advanced techniques introduced in the healthcare domain to measure biometric data directly from the human brain and decode it to gain insights into the physical and mental health of individuals [3]. This data can then be utilized to form interactions with the physical world or achieve smart control by designing equipment such as a smart wheelchair, a prosthetic arm, or smart cursor control. This interaction of the human brain with external peripheral devices is termed as Brain-computer

The associate editor coordinating the review of this manuscript and approving it for publication was Jolanta Mizera-Pietraszko.

interface (BCI) which is a very prominent trend in the medical domain as it allows individuals possessing motor disabilities to interact with devices using their motor imagery (MI) signals thus enhancing the quality of life [4], [5]. However, BCI for MI classification is not a straightforward application as the MI signals are complicated with high dimensionality. Therefore several researchers endeavor to explore different machine learning or deep learning approaches to decode the complex MI data [2], [4].

Typically, MI features can be categorized into spectral, spatial, and temporal features [6]. Spectral features contain either frequency-domain or time-frequency domain features. Techniques like power spectral density (PSD) [7] or fast Fourier transform (FFT) [8] can be used for spectral feature extraction. Spatial features are extracted from specific electrode positions on the scalp by utilizing techniques such as common spatial pattern (CSP) [9] and its derivatives including filter-bank CSP [10]. Temporal features, on the other hand, refer to features at various time points in the time domain such as skewness, variance, or mean [11]. Following the feature extraction, different classifiers such as support vector machine (SVM) [12], linear discriminant analysis (LDA) [13], or extreme learning machine (ELM) [14] is commonly utilized to classify the MI features into the respective MI tasks.

Numerous studies have examined the efficiency of MI classification using signal processing and machine learning techniques. Alam et al. [6] utilize PSD to extract features from BCI Competition IV-2b and use LDA to classify those features and demonstrate a Cohen's kappa accuracy of 0.61 in subject-specific applications. A novel ensemble learning framework based on CSP is proposed in [15] by utilizing Thikonov regularization technique. The authors have utilized five MI datasets and claim an average accuracy of 85% across those datasets. Similarly, in [16], a brain-to-unarmed aerial vehicle (UAV) communication is established using a stacked ensemble CSP algorithm. Annaby et al. [17] used digraph Fourier transform for feature extraction and classified the features using extreme learning machine (ELM) obtaining an average classification accuracy of 96.58% on dataset 1a of BCI competition 2003. Moreover in [18], a matrix encoded slicing method is used to classify BCI Competition III-IVa MI tasks with an average accuracy of 99%.

Although a combination of signal processing and machine learning algorithms has shown considerable success in MI classification, however, these methods have their own demerits such as the easy influence of artifacts, sensitivity to their specific parameters, and environmental noise [19]. Therefore, in the last decade, deep learning methods have been explored to eliminate extensive pre-processing and feature extraction stages. Deep learning techniques automatically learn complex patterns from the raw MI data using deep convolutional architecture and have great scalability with training data size [20]. Several researches have explored the DL techniques to enhance MI classification and have achieved remarkable performance [3], [21]. DL models consist of multiple layers of interconnected nodes to process the MI data and classify it into MI tasks. The learning of a DL model is based on the minimization of the loss function by adjusting the weights and biases of the connected nodes. In [22], a combination of convolution neural network (CNN) and long short-term memory (LSTM) is utilized for MI classification. Similarly, in [23], a pre-trained Alex net is used for EEG classification using transfer learning, and 98% accuracy is reported on BCI III-IVa dataset. EEGNet is introduced by Lawhern et al. [24] to classify MI signals using depth-wise and separable convolution layers. The depth-wise and separable convolutions work efficiently to extract features while also reducing the parameters by replacing the standard convolution layer with a separable convolution layer. Moreover, ShallowNet is proposed to extract features using only two convolution layers followed by an average pooling layer whereas DeepConvNet is designed to extract complex features using a deep convolutional architecture [25]. However, finding the optimal weights and biases is often challenging and can lead to sub-optimal performance.

In the past few years, transformers have initiated a new wave of feature extraction which has entirely revolutionized the deep learning domain by introducing the self-attention and cross-attention mechanism [26]. The key innovation of this technology relies on its ability to weigh the different points in the input sequence differently rather than relying on traditional sequential processing. This helps the model to learn better features by understanding the relations between different points in the input sequence. The remarkable success of transformers in the natural language processing (NLP) domain has drawn sufficient research attention to utilize this in the EEG MI classification domain [11], [27]. For instance, TACNet [28] is proposed that utilize the self-attention mechanism to extract task-related features and electrode channel interrelation. A noteworthy accuracy improvement of 11.4% compared to the state-of-the-art on BCI Competition IV-2a is reported by the authors. Xinzhi et al. [4] have also used the attention mechanism for temporal features learning and have quoted an accuracy of 79.48% on BCI IV-2a dataset. Moreover, a learnable channel selection approach is designed in [29] using channel attention to enhance MI EEG decoding accuracy.

However, most existing studies only focus on one domain of features such as [30] is focused on temporal features extraction whereas [31] focuses on spatial relations. Nevertheless, an in-depth analysis and classification of the signal requires features extracted from spectral, temporal as well as spatial domains. Moreover, the existing studies are mostly tailored to subject-specific classification whereas such classification techniques are often not reliable when testing in real-life situations due to weak generalization capability. To address such issues, we present TriNet which provides a unique concatenation of features from three significant domains to classify MI signals. Our approach utilizes FFT to extract spectral features, a spatial transformer

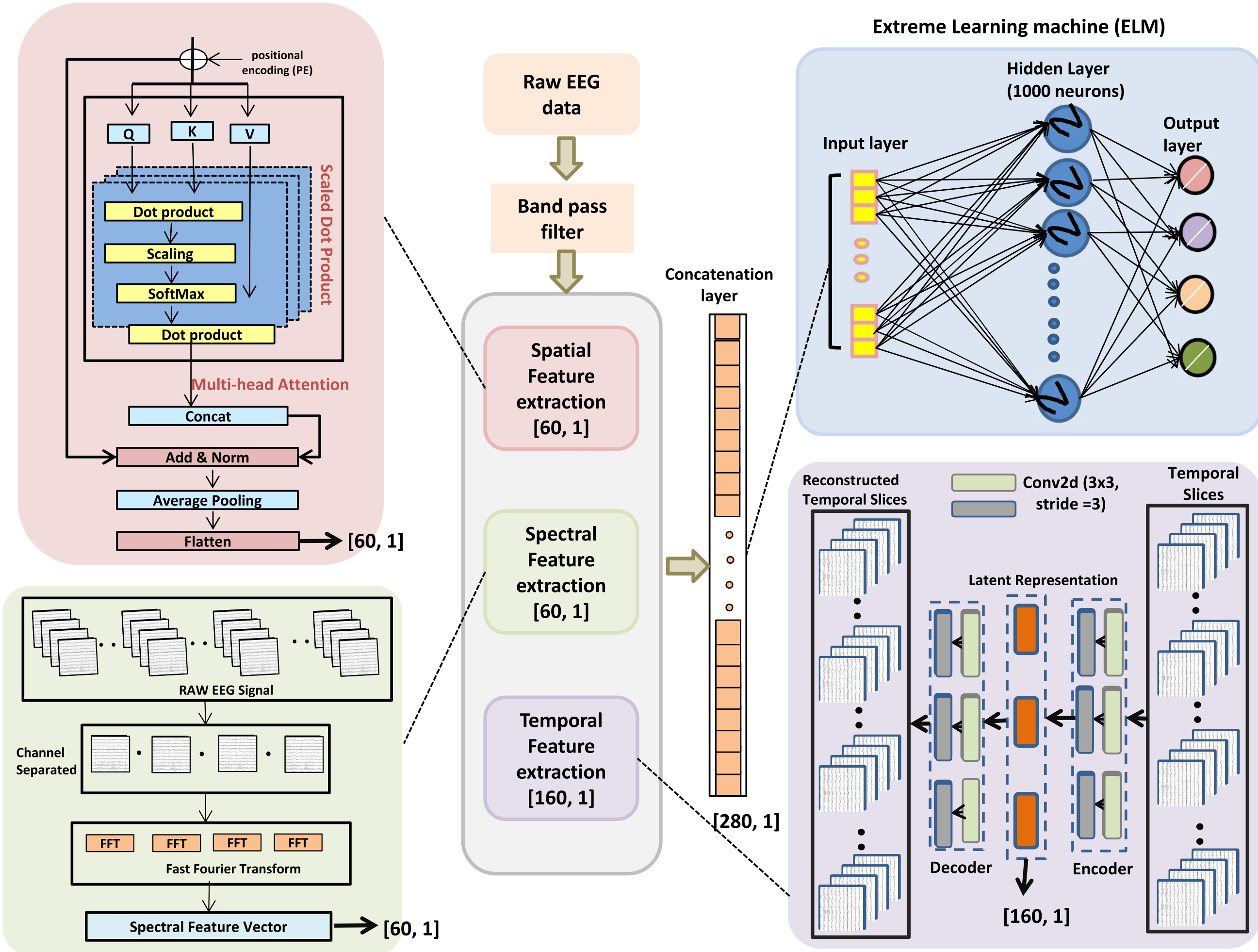

FIGURE 1. Detailed architecture of the proposed TriNet. Green block represents the spectral feature extraction module, pink block shows the spatial feature extraction module and purple unit represents encoder-decode module, blue blocks represents ELM-based classification.

to extract spatial features, and an encoder-decoder structure to extract temporal features. Moreover, the extracted features are also classified using ELM which has better generalization capability compared to the traditional feed-forward layer. To reflect the core structure of our framework, we hereby name it TriNet which emphasizes its architecture consisting of three distinct parallel branches. TriNet advances the field of MI classification in BCI in the following four ways:

1) A novel feature integration approach is proposed that leverages features from three distinct domains: spectral, spatial and temporal to provide a holistic representation of the MI signal.
2) The feature classification is also optimized by using ELM instead of a feed forward layer to achieve better generalization performance.
3) A subject-independent approach is designed for MI classification to ensure robustness and achieve scalability across different individuals.
4) TriNet is tested on both two and four-class MI datasets and an average accuracy improvement of 8 to 10% compared to existing methods is observed.

The rest of this paper is organized in the following manner: section II explains the working of all the required modules of TriNet. Section III describes the pertinent details of the database as well as the experimental setup and also presents results and key analysis of TriNet. Section IV enlists discussions on the number of parameters as well as the ablation studies. Section V concludes our study.

II. METHODOLOGY

TriNet aims to present a unique classification approach for MI classification in BCI by employing spectral, spatial, and temporal feature extraction to accurately interpret EEG signals. The detailed framework of TriNet is shown in Fig 1. This method involves preprocessing EEG signals first by running them through a band-pass filter. This removes unnecessary noise and isolates the 4-30 Hz frequency range, which is crucial for activities involving MI, especially for capturing the mu and beta rhythms associated with motor action. Following filtering, three separate parts handle the filtered EEG signal in parallel: an encoder-decoder structure, a self-attention module, and FFT. The signal from the time domain is converted into complex frequency domain using FFT, enabling extraction of the spectral characteristics which shows the power distribution over pertinent frequency bands. Concurrently, the self-attention module is utilized to record spatial dependences across various EEG channels, augmenting the system's capability to simulate inter-regional interactions, specifically in motor-related domains. This module assigns weights to the interactions between various channels to help identify spatial features. The temporal dynamics of the EEG signals are modeled concurrently using the encoder-decoder structure, in which the encoder records the temporal dependencies in the time series and the decoder reconstructs the signal, highlighting important temporal aspects.

Following the processing of the signals by these three components (FFT, self-attention, and the encoder-decoder), the outputs, which include spectral, spatial, and temporal information, are merged to create an extensive and enhanced feature set. Then, the comprehensive data set is fed into the ELM, a neural network that is single-layered and feed-forward. It is very effective in the classification as compared to the conventional feed-forward neural network due to its better generalization capabilities so it is considered over the other classifiers. Each module in TriNet is explicitly described in the preceding sections.

A. FILTERING

Preprocessing is a crucial step in analyzing EEG signals, especially in BCI systems. EEG signals are often contaminated by various artifacts and environmental disturbances. Therefore, removing these unwanted disturbances is essential to ensure that the signal accurately represents the MI task. Band-pass filtering [42] is commonly used to isolate the signal in the desired frequency range.

A band-pass filter works by allowing only a specific range of frequencies to pass while blocking others. For MI tasks, brain oscillations between 4 to 30 Hz are considered significant, as these frequencies are associated with motor tasks. The mathematical representation of a band-pass filter in the frequency domain is given as:

$$H(f) = \begin{cases} 1, & \text{if } f_1 \leq f \leq f_2 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $H(f)$ represents the transfer function of the signal in the frequency domain, and f_1 and f_2 are the cut-off low and high frequencies, respectively. In the time domain, the band-pass filter can be expressed as a convolution between the input signal $x(t)$ and the impulse response $h(t)$ of the filter:

$$x_{\text{filtered}}(t) = x(t) * h(t) \tag{2}$$

B. SPECTRAL FEATURES EXTRACTION USING FFT

FFT is commonly employed in signal processing to transform the signal from the time-domain to the complex frequency domain. After passing through a band-pass filter to remove unwanted frequencies and noise, the spectral characteristics of the EEG data, which are crucial for understanding and classifying brain activity, especially in tasks involving MI, are rendered visible by this conversion. This transformation can help identify and analyze brain patterns unique to a frequency by analyzing the FFT output, which provides a comprehensive view of the power distribution across several frequencies.

The main objective of applying FFT to EEG data is to extract spectral features, which indicate the power distribution of the signal at different frequencies. When the EEG signal is represented in the time domain, it exhibits voltage changes across time. FFT is used to break down the signal into its frequency components, revealing the amount of power or energy that is contained in each frequency band. Mathematically, FFT can be represented as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \tag{3}$$

where $X[k]$ represents the k-th component of the frequency transform, n represents the n-th sample, and N denotes the total number of samples, whereas j is the imaginary unit. The output of FFT consists of several frequency bins, each representing a distinct frequency component of the original time-domain signal. When the EEG signal is sampled at a frequency f_s and contains N samples, the FFT yields N complex values, each of which corresponds to a distinct frequency between 0 and $f_s/2$ Hz. This range is determined by the Nyquist Theorem, which stipulates that the maximum frequency that may be resolved is half the sampling rate, f_s.

C. SPATIAL FEATURES EXTRACTION USING SELF-ATTENTION

Once the band-pass filter has preprocessed the EEG data to isolate the important frequency range (typically 4-30 Hz), spatial feature extraction is necessary to capture the dependencies across the distinct EEG channels. In this particular circumstance, modeling these spatial relationships is critically dependent on the self-attention process. With the use of self-attention, the model may dynamically focus on several EEG channels, highlighting the signal components that are particularly important for MI tasks. This makes it possible for the network to record interactions and inter-channel dependencies, which are essential for comprehending brain dynamics.

Multiple electrodes are used to record EEG data, and each electrode indicates neural activity from a different area of the brain. Neural signals, however, are not localized; during activities like MI, several brain regions frequently interact in intricate ways. It is crucial to represent the connections between channels, or electrodes, to extract spatial properties that characterize the communication between various brain regions and capture these interactions.

TriNet employs the self-attention mechanism to extract those intricate spatial dependencies in the EEG data. Self-attention mechanism allows the network to prioritize certain parts of the input over others by computing a weighted combination of input properties. The self-attention mechanism can be applied to the channels (spatial dimension) of the EEG signal in the context of EEG spatial feature extraction.

The self-attention mechanism works by computing the associations between the channels by generating Query, Key, and Value matrices for each given input. The attention weights that establish each channel's relative relevance to the others are calculated using these matrices. The process of self-attention can be described mathematically as follows:

Key (K), Query (Q), and Value (V): Initially, the self-attention mechanism projects the input X into three distinct spaces: values V, keys K, and queries Q. These projections, which represent various facets of the input features, are acquired through transformations represented as:

$$Q = W_Q X \tag{4}$$

$$K = W_K X \tag{5}$$

$$V = W_V X \tag{6}$$

Attention Weights: The scaled dot product between a channel's query and its key represents the attention score between two channels. The model's appropriate level of emphasis on each channel in respect to the others is determined by this operation:

$$\text{Attention}(Q, K) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \tag{7}$$

These attention values allow the network to describe intricate spatial interactions across the scalp by indicating how much attention each EEG channel should devote to the others. The final output is computed by taking a dot product of the attention matrix with the value matrix:

$$\text{Output} = \text{Attention}(Q, K)V \tag{8}$$

Self-attention dynamically modifies the focus on several EEG channels according to the task at hand, in contrast to conventional approaches that handle each channel independently. As a result, the channels that are more important for categorizing tasks involving MI will receive greater attention from the network. When dealing with high-dimensional inputs such as EEG data, where the number of channels varies between datasets or applications, the self-attention mechanism scales effectively.

D. TEMPORAL FEATURES EXTRACTION USING ENCODER-DECODER

Upon undergoing bandpass filter preprocessing to identify the pertinent frequency range from EEG signals, temporal feature extraction becomes crucial in comprehending the dynamic patterns over time. Because EEG records intrinsically time-dependent brain activity, MI tasks generate signals that change over time. These temporal correlations are well captured by the encoder-decoder architecture, which offers a thorough depiction of the alterations in brain activity across time. For sequence-to-sequence operations, where the objective is to map a sequence of inputs to a sequence of outputs, the encoder-decoder architecture is frequently employed. The architecture is shown in Fig 2 and is used to record the temporal structure of the input signals in the context of EEG signal processing.

The temporal feature extraction module consists of two core components: encoder and decoder, as illustrated in Table 1. Initially, the EEG signals are passed to the encoder component of the architecture, which compresses the input EEG signal into a compressed lower-dimensional representation that contains the temporal features. The encoder consists of two convolutional layers where the first layer extracts the local features from the EEG data and the second layer further processes those features to enhance the feature representation. Afterwards, a linear layer flattens the features and summarizes the input information. The encoding function can be represented as:

$$Z = f_{\text{encoder}}(X_{\text{filtered}}) \tag{9}$$

Here, Z is the latent representation that captures the crucial temporal characteristics of the signals, focusing on the characteristics that are mainly related to the MI task. f_{encoder} is the encoding function that represents the convolution and flattening operation.

The latent representation is then fed into the decoder, whose main function is to reconstruct the original filtered signal. The decoder structure involves a series of transpose convolution layers to effectively reverse the encoder operation, thus reconstructing the original signal from the latent representation. The decoding process can be expressed as:

$$\hat{X} = f_{\text{decoder}}(Z) \tag{10}$$

where, $\hat{X}$ is the reconstructed signal produced by the decoder block, whereas f_{decoder} is the decoding function. During the training of the encoder-decoder module, the encoder learns to produce a better compressed representation of the EEG signal, and the decoder aims to reconstruct the signal to closely resemble the original EEG signal.

E. ELM FOR CLASSIFICATION OF MI STATES

The last stage of the TriNet is to feed all the extracted features, including the spectral features extracted using the Fast Fourier Transform, the spatial features extracted using the Self-attention method, and the temporal features extracted using the Encoder-decoder, into the fast and accurate ELM for the classification of MI states. The data fed into the ELM is a comprehensive and large feature set, which is a combination of all three extracted features. The choice of ELM over other classifiers is due to its proven superiority in classification and speed. Its non-iterative learning technique and efficiency in handling large datasets makes it a good fit for EEG data classification.

The first layer in the ELM is the input layer that receives the aggregate feature set, where each feature represents a complex pattern of the EEG data. The next is the hidden layer,

TABLE 1. Architecture of the encoder-decoder block.

Layer (Type)	Output Shape	Kernel Size	Stride	Padding	Output Padding
Input	(N, 1, 500, C)	–	–	–	–
Conv2d (C1, C2)	(N, 2, 167, C)	3	3	1	–
Conv2d (C2, C1)	(N, 1, 56, C)	3	3	1	–
Linear (1×56×1, 160)	(N, 160)	–	–	–	–
Encoder Output	**(N, 160)**	–	–	–	–
Linear (160, 1×56×1)	(N, 56)	–	–	–	–
ConvTranspose2d (C1, C2)	(N, 2, 167, 1)	3	3	1	(1, 0)
ConvTranspose2d (C2, C3)	(N, C, 500, 1)	3	3	1	(1, 0)
Decoder Output	**(N, C, 500, 1)**	–	–	–	–

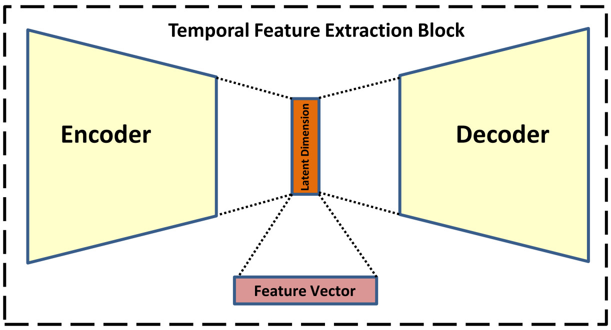

FIGURE 2. Encoder-decoder architecture for temporal feature extaction.

which does not require any adjustment during training. The final layer is the output layer, which generates a class label for the anticipated MI, such as the left or right movement of the hands.

If the input data from the large feature set containing N samples is defined as:

$$X = \left[x_1, x_2, x_3, \ldots, x_N\right] \tag{11}$$

Given this input, the ELM performs the following operations:

Initially, random assignments are made to the input weights and biases. After receiving input from the feature set, each neuron in the hidden layer performs a nonlinear transformation using an activation function:

$$h(X) = f(XW + b) \tag{12}$$

where X represents the input, W represents the weight matrix, and b denotes the bias added to each neuron.

The final step is to compute the output weights $\beta \in \mathbb{R}^{L\times C}$, where C is the number of MI classes (e.g., left hand, right hand) and L is the number of neurons in the hidden layer. The output weights β are learned by minimizing the difference between the output of the hidden layer and the actual class labels.

The relationship between the hidden layer output H and the output labels Y is expressed as:

$$H\beta = Y \tag{13}$$

The Moore-Penrose pseudo-inverse is used to calculate the analytical output weights β in ELM:

$$\beta = H^{\dagger}Y \tag{14}$$

where $H^{\dagger}$ is the Moore-Penrose pseudo-inverse of the matrix H, defined as:

$$H^{\dagger} = \left(H^T H\right)^{-1} H^T \tag{15}$$

By solving for β, the model determines the optimal weights that map the hidden layer outputs to the correct MI labels.

III. EXPERIMENTAL SETUP AND RESULTS

A. DATABASE

To demonstrate the effectiveness of TriNet, we have utilized two benchmark datasets from BCI Competition IV, which were recorded by Graz University of Technology. Both datasets are publicly available and extensively used for evaluating MI classification frameworks. The MI data acquisition setup can be seen in Fig. 3. The details of these datasets are presented in the subsequent sections:

BCI Competition IV Dataset 2a:

BCI Competition IV-2a is a four-class classification dataset in which four MI tasks are performed: movement of the left hand, right hand, foot, and tongue. The data was recorded for nine subjects at a frequency of 250 Hz using 22 electrodes placed on the subject's scalp. Two sessions were conducted for recording: one for training and one for testing. The time duration taken for each trial is between 2 to 6 seconds, resulting in the shape of a single trial being 1000 × 22.

BCI Competition IV Dataset 2b:

BCI Competition IV-2b is a binary-class dataset where subjects are required to perform two MI tasks: movement of the left and right hands. The sampling frequency for this dataset acquisition was also set at 250 Hz. However, in this case, 3 electrodes were used to collect data from nine healthy participants. The data was recorded in five sessions, with data from three sessions serving as the training set, while the remaining two sessions were used for testing. The signal duration is between 3 to 7 seconds, making the shape of a single trial in this dataset 1000 × 3.

TABLE 2. TriNet performance comparison on BCI IV-2a dataset.

Method	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg ± Std	Avg. K-value	p-value
Shallow ConvNet [26]	80.20	54.86	92.01	59.03	66.67	54.17	87.50	79.51	82.29	72.92 ± 14.5	0.63	$p < 0.01$
Deep ConvNet [26]	80.90	52.08	84.72	71.18	70.49	55.56	69.10	81.94	81.94	71.99 ± 11.8	0.62	$p < 0.01$
CP-MixedNet [34]	74.65	53.47	73.26	70.14	67.36	48.96	74.31	72.92	69.44	67.17 ± 9.4	0.56	$p < 0.01$
ATCNet [35]	88.61	73.85	**97.44**	81.58	80.80	73.95	93.50	**90.04**	89.77	85.50 ± 7.90	0.80	$p = 0.07$
EEGNET [25]	89.32	63.25	94.87	67.54	75.00	60.93	89.89	87.82	78.90	81.44 ± 11.93	0.71	$p < 0.05$
EEG-NeX [36]	85.05	71.02	96.34	**85.09**	81.52	63.72	90.61	83.39	84.09	82.32 ± 9.18	0.76	$p < 0.05$
MBEEGSE [37]	**90.39**	65.37	95.60	75.88	77.54	61.40	93.14	88.56	85.98	81.54 ± 11.52	0.75	$p < 0.05$
TS-SEFFNet [38]	82.29	49.79	87.57	71.74	70.83	63.75	82.92	81.53	81.94	74.71 ± 0.12	0.75	$p < 0.01$
Proposed TriNet	90.10	**75.30**	95.40	84.00	**86.00**	**79.00**	**94.80**	89.60	**91.80**	**87.30 ± 0.15**	**0.82**	− − −

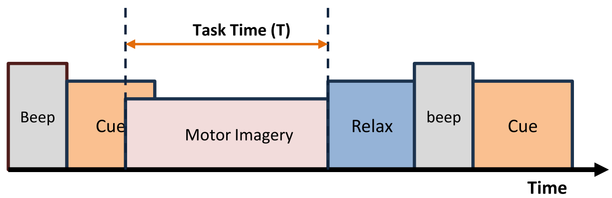

FIGURE 3. Motor imagery data acquisition setup.

B. IMPLEMENTATION SETUP

All experimentation for TriNet is performed on a 12th generation Intel Core i7 with 32 GB RAM and an NVIDIA RTX-4060 graphic card. The simulations are done on Visual Studio Code and MATLAB. The EEG trials obtained from the aforementioned datasets have a shape of 1000×22 which is further split to make each trial of 500×22 shape to increase the data size. To obtain the relevant EEG signals corresponding to MI tasks, we have utilized a band-pass filter ranging between 3 to 40 Hz to filter out the desired signals.

Next, the filtered signal is simultaneously fed to three parallel feature extraction branches to extract spectral, spatial, and temporal features. The spectral features are extracted using the FFT, which transforms the EEG signal from the time domain to the complex frequency domain. This helps to identify the key components of frequency associated with the given MI tasks. In our study, we have selected the top 60 components from FFT, so the output shape from this branch becomes (60, 1).

In parallel, a spatial transformer employs a self-attention mechanism to effectively capture spatial dependencies within the EEG signals, resulting in an output vector of shape (60,1). This self-attention mechanism allows the model to weigh the importance of different spatial features, enhancing the extraction of relevant spatial patterns associated with MI tasks. For temporal feature extraction, we implement an Encoder-Decoder architecture. During the training process, the model learns to map the input features to a latent representation. Once the training is complete, the decoder component is removed, leaving only the encoder. The encoder's output serves as the latent representation of the temporal features, yielding a vector with 160 features that encapsulate the temporal dynamics of the EEG signals.

Ultimately, the extracted spectral, spatial, and temporal features are concatenated into a single vector of shape (280,1). This comprehensive feature vector is then input into an ELM with 1000 hidden neurons. The ELM is utilized for its rapid training capabilities and efficient handling of high-dimensional data, facilitating the classification of MI tasks based on the rich feature representation derived from the EEG signals.

C. PERFORMANCE EVALUATION METRICS

In order to evaluate TriNet, we used three primary performance evaluation metrics, namely accuracy (A), kappa value (K), and F1 score. Accuracy can be defined as a ratio of correctly predicted instances to the total number of instances taken into account. Accuracy can provide a straightforward indication of the TriNet performance; however, it is mostly considered with the assumption that data is perfectly balanced. Accuracy can mathematically be defined as:

$$\mathrm{A} = \frac{Truepos. + Trueneg.}{Truepos. + Trueneg. + Falsepos. + Falseneg} \tag{16}$$

where *pos*. represents positive outcomes whereas *neg*. represents negative outcomes. In the case of imbalanced datasets where one class significantly outnumbers the other class, accuracy alone cannot adequately reflect the performance of the designed model. Hence, we utilize the F1 score, defined as:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

F1 score provides a harmonic mean between recall and precision, where precision is the ratio of true pos. to the sum of true pos. and false pos., and recall is the ratio of true pos. to the sum of true pos. and false neg. Precision and recall are defined as:

$$\text{Precision} = \frac{TruePos.}{TruePos. + FalsePos.} \tag{18}$$

$$\text{Recall} = \frac{TruePos.}{TruePos. + FalseNeg.} \tag{19}$$

Additionally, we have also utilized the Cohen's Kappa coefficient for our model evaluation. Kappa values assess how well the predicted and actual classifications agree with each other, determining the degree of reliability in the overall classification performance. Kappa can be defined as:

$$\text{Kappa} = \frac{P_o - P_e}{1 - P_e} \tag{20}$$

TABLE 3. TriNet performance comparison on BCI IV-2b dataset.

Method	B0103T	B0203T	B0303T	B0403T	B0503T	B0603T	B0703T	B0803T	B0903T	Avg ± Std	p-value
SGRM [39]	76.30	56.00	49.00	98.20	91.10	74.80	88.30	85.40	84.90	78.78 ± 16.3	$p < 0.01$
CSP with clsSRC2 [40]	71.88	59.64	57.19	95.63	92.50	82.50	73.75	90.00	85.94	78.20 ± 14.0	$p < 0.01$
RCSP with clsSRC [41]	71.25	58.21	56.56	95.94	91.56	82.19	74.06	90.94	85.62	78.48 ± 14.4	$p < 0.01$
RCSP with clsSRC2 [40]	77.63	56.79	58.44	96.25	91.25	81.25	73.44	90.63	86.56	78.36 ± 14.4	$p < 0.01$
PSD-SVM [42]	73.12	61.78	58.43	93.75	87.50	82.81	76.87	92.19	87.50	79.33 ± 7.31	$p < 0.01$
FBCSP-RLDA [43]	80.70	67.00	57.00	88.50	83.10	75.60	78.90	73.50	73.60	75.36 ± 9.26	$p < 0.01$
DRDA [46]	81.37	62.86	63.63	95.94	93.56	88.19	85.00	95.25	90.00	83.98 ± 5.13	$p < 0.05$
Conformer [45]	82.50	65.71	63.75	**98.44**	86.56	90.31	87.81	**94.38**	92.19	84.63 ± 4.86	$p < 0.05$
Proposed TriNet	**93.65**	**90.43**	**91.70**	95.43	**96.09**	**92.26**	**90.48**	88.67	**95.08**	**92.64 ± 2.31**	− − −

where P_o is the observed agreement and P_e is the expected agreement by chance. Kappa is ranged between -1 to 1, where 1 indicates that predicted and actual classifications are in perfect agreement.

To demonstrate the statistical significance of the proposed TriNet, Wilcoxon rank test with a significance level of 0.05 is used. The null hypothesis states that there is no significant difference between TriNet and the existing methods whereas the alternate hypothesis suggests that TriNet has a statistical significance over the existing and state-of-the-art methods. Tables 2 and 3 demonstrate the statistical test results with $p < 0.05$ indicating significant difference with a 95% confidence level.

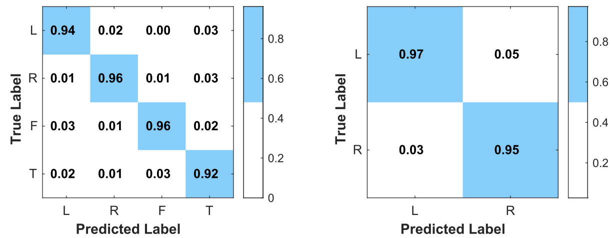

FIGURE 4. Normalized confusion matrices on four-class (BCI IV-2a) and two-class (BCI IV-2b) datasets.

D. RESULTS AND ANALYSIS ON SUBJECT-SPECIFIC CLASSIFICATION

This section enlists the performance comparison of TriNet with various state-of-the-art and recently developed methods on the basis of accuracy and kappa value. Table 2 shows the comparison in terms of individual subject performance and the average performance on BCI IV-2a dataset. The average accuracy of TriNet across all subjects is 87.30% which is a considerable improvement over the state-of-the-art and various existing methods.

First, we compare TriNet with Shallownet, a shallow deep learning architecture with two convolution layers followed by an average pooling layer. The average accuracy of Shallownet across all subjects is 72.92% with a high std. of 14.5 whereas TriNet, with an average accuracy of 87.30%, has a lead of around 15% compared to Shallownet with a very low std which demonstrates a statistical improvement with $p < 0.01$. In comparison to DeepConvNet, which is a deeper architecture compared to Shallownet, a similar pattern is observed, with an average gain of more than 15% and a significant reduction in the number of parameters. Moreover, in terms of kappa value comparison, Shallownet and DeepConvNet have kappa values of 0.63 and 0.62, respectively, which are far lower than 0.82 of TriNet. Next, we compare it with CP-MixedNet, which extracts temporal features using a multi-scale approach. TriNet has achieved superior performance on all instances, with a remarkable average accuracy improvement of 20% ($p < 0.01$) and far lesser computational load. In comparison to ATCNet, which uses a novel physics-informed technique to extract interpretable temporal features from EEG data, our TriNet has achieved comparable and better results. An average accuracy gain of 2% is observed with better performance on seven subjects. On subjects 3 and 8, ATCNet has a marginal accuracy gain, however, except for the weak results on these two subjects, TriNet has a clear lead and still has a statistical improvement with $p < 0.05$. The std. of ATCNet is also significantly higher compared to TriNet which emphasizes more stable results of TriNet.

Moreover, to compare with EEGNet, which uses separable and depth-wise convolution layers to extract features, TriNet shows superior performance on all subjects. Especially on subjects 4, 5, and 6, the gain in accuracy is substantial. Further, we compare our method with EEG-Nex, which uses a pure ConvNet-based architecture. It can be seen that EEG-Nex only performed marginally better on subjects 3 and 4, however, for rest of the subjects, TriNet has a considerably higher accuracy with $p < 0.05$. Moreover, EEG-Nex has a highly complex architecture compared to the proposed TriNet. Similarly, MBEEGSE, which uses a multi-branch architecture with a squeeze and excitation block, has an average accuracy of 81.54% and a high std of 11.52, whereas TriNet has outperformed with a performance improvement of 6% and a very low std. Finally, we compare our proposed TriNet with TS-SEFT-Net, which uses multiple convolutional blocks for multi-domain features. TriNet has achieved significantly better results on all subjects, and an average accuracy improvement of 13% is observed.

In terms of kappa value, TriNet's kappa values average at 0.82, surpassing all the comparable models, which demonstrates a strong similarity between our predicted and actual classification results. In comparison to the competitive models, ATCNet has the highest kappa value of 0.80, which is still significantly lower than TriNet. Overall, while several models struggle to maintain performance, TriNet shows robustness and stability with a minimum accuracy of 75%. For instance, while several models exhibit a sharp decline in accuracy on subject 2, TriNet, with an accuracy of 75.3%, remains relatively robust, showing that it is less susceptible to inter-subject variations.

Next, we conduct an analysis of the classification accuracy of TriNet on BCI Competition IV Dataset 2b in comparison with various established approaches, including SGRM, CSP with clsSRC2, RCSP with clsSRC, RCSP with clsSRC2, PSD-SVM, and FBCSP-RLDA, across multiple subjects as shown in table 3. The experimental results reveal that TriNet has surpassed all the existing models with significant improvement in accuracy and $p < 0.01$ in most cases. TriNet has an average accuracy of 92.64%, whereas the best-performing model, Conformer, has an accuracy of 84.63%, which shows a significant accuracy gain of 8%.

On an individual subject level, TriNet has outperformed other methods on all subjects except for EEG-Conformer, which exhibits slightly better performance on subjects 4 and 8 compared to TriNet. The stability of TriNet is evident when compared on the basis of performance fluctuations at the individual subject level. For instance, SGRM attains the lowest accuracy of 49% on subject 3 and the highest accuracy of 98%, reflecting strong inconsistency. A similar trend is observed in other studies, where some subjects perform exceptionally poor. Interestingly, EEG-Conformer also experienced an accuracy drop on subjects 2 and 3, but TriNet maintained good performance on all subjects, including subject 2, which demonstrates the effectiveness of our approach. Figure 4 shows the normalized confusion matrices of TriNet on the two datasets where the x-axis represents the predictions by TriNet whereas the y-axis represents the ground truth. Moreover, Fig. 6 represents the feature discrimination capability of TriNet where it can be seen that TriNet has been able to discriminate features efficiently in both two and four-class MI tasks.

E. RESULTS AND ANALYSIS ON SUBJECT-INDEPENDENT CLASSIFICATION

EEG data is inherently prone to artifacts, and individual subject differences can greatly impair EEG decoding accuracy when testing on unseen subject data. This occurs because, for the same electrode positioning, different brain regions are activated for different subjects. Thus, most studies focus on subject-specific classification accuracies but do not pay much attention to subject-independent classification. However, for a fair analysis of TriNet, we evaluate its robustness in both subject-specific and subject-independent setups.

Moreover, since the datasets we use are not well-balanced, F1-score results are also considered in our evaluation. We have implemented the renowned LOSO (Leave-One-Subject-Out) [45] technique, which provides insights into TriNet's performance across multiple folds. In each fold, the data from one subject is held out for testing, while the data from the rest of the subjects is used as the training set. This method ensures that the model is tested on completely unseen data and serves as a great indicator of the EEG decoding capability of TriNet.

TABLE 4. LOSO technique results on BCI IV-2a dataset.

Dataset	Fold	Accuracy	F1-Score
BCI-IV 2a	V1	60.0	0.59
	V2	69.3	0.67
	V3	69.5	0.68
	V4	55.2	0.55
	V5	69.5	0.64
	V6	65.4	0.64
	V7	58.2	0.57
	V8	**70.2**	**0.68**
	V9	57.8	0.56

Table 4 shows the classification accuracy and F1 score of TriNet in the subject-independent setup on the BCI IV-2a dataset. The results indicate that, despite the inherent complexities associated with subject variability, TriNet still showcases commendable classification performance on several instances, underscoring its potential to generalize for effective MI decoding. For instance, TriNet achieves an accuracy of 70.2% in fold V8, reflecting its strong capability to generalize to unseen data. Similarly, in folds V2, V3, V5, and V6, TriNet ensures robust performance on new subjects' data. Even on the folds where performance is not as high, such as fold V4 and V7, the respective accuracies of 55.2% and 58.2% are still notable, considering the strict testing conditions implied in the LOSO technique. On average, TriNet demonstrates robustness with an average accuracy of 63.9% on the BCI IV-2a dataset. Moreover, the F1 score across all folds demonstrates that TriNet is effectively balancing precision and recall, highlighting its ability to capture intricate patterns of EEG data for MI classification.

Further, Table 5 compares the average classification performance and F1 score of TriNet in both subject-specific and subject-independent scenarios on the BCI IV-2b dataset. First, TriNet shows an exceptional classification accuracy of 92.6% and an F1 score of 0.92 in subject-specific classification, as demonstrated in Table 3. This noteworthy performance highlights the effectiveness of integrating multi-domain features in capturing the intricate dependencies of EEG data. On the contrary, the classification accuracy and F1 score of 78.6% and 0.78, respectively, in subject-independent classification reflects its ability to maintain satisfactory performance even on unseen data.

IV. DISCUSSIONS AND CONSIDERATIONS

A. ABLATION EXPERIMENTS

In order to validate the significance of features from the three distinct temporal, spectral, and spatial domains, an ablation study is conducted on the BCI IV-2b dataset. In this ablation

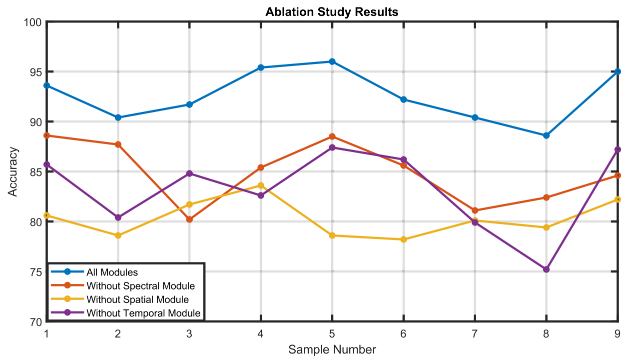

FIGURE 5. Ablation study performed on TriNet.

study, the model's performance is evaluated with and without the three feature extraction modules. Figure 5 showcases the ablation results and the significance of each module in attaining the desired classification performance. From the figure, it can be seen that TriNet model, which incorporates all three feature extraction blocks, shows an impressive accuracy of 92.6%. This accuracy serves as the benchmark to compare the performance degradation when individual feature extraction blocks are removed. First, the encoder-decoder block is removed, which dropped the average accuracy to 83.27%, indicating a performance degradation of nearly 7.9%. This decrease in accuracy highlights the crucial role that temporal dependencies play in capturing the inherent patterns of the EEG data. The degradation in performance is most pronounced in subject 4, dropping from 95.4% to 82.6%, a considerable decline of 12.8%. Other subjects, such as subjects 2 and 5, also experience significant accuracy drops of 10% and 8.6%, respectively, emphasizing the importance of the temporal feature extraction block. Subjects 1, 3, and 9 showed comparatively lower performance degradation, but their performance degradation is still 5 to 6%, which is a significant performance loss.

Next, when the spatial feature extraction block is removed, an even greater decline in accuracy is observed, with the average accuracy dropping to 80.33%. All subjects have shown significant performance degradation, with nearly 8-15% accuracy drops. This shows that the performance of TriNet strongly relies on the spatial feature extraction block.

Moreover, when FFT is removed and there are no spectral features, the accuracy falls to 84.9%. Although the drop is not as high as when the temporal and spatial feature extraction blocks were removed, a drop from 92.6% to 84.9% is still notable. Some subjects, such as subject 2, experienced minor performance degradation, whereas in other subjects, the accuracy was considerably reduced, indicating the significance of spectral features. Overall, the ablation experiments reveal that the spatial feature extraction block is the most critical in ensuring high classification accuracy across all subjects. Additionally, the ablation study demonstrates that TriNet is intricately designed, with each feature extraction block having its own strength, and removal of any of these blocks can significantly impair the performance of EEG decoding.

TABLE 5. Accuracy and F1-Score for subject-dependent and subject-independent methods on BCI IV-2b.

Method	Accuracy	F1-Score
Subject dependent	**92.6 (std 3.0)**	**0.92 (+0.04)**
Subject independent	78.6	0.781

TABLE 6. TriNet parameter comparison with existing methods.

Method	Number of Parameters (Million)
Shallow ConvNet [26]	0.047
Deep ConvNet [26]	0.284
CP-MixedNet [34]	0.836
TS-SEFFNet [38]	0.282
TriNet	**0.018**

B. NUMBER OF PARAMETERS

Table 6 displays the number of parameters of TriNet and compares it to other existing methods. The number of parameters indicates the computational cost of the model, with fewer parameters suggesting that the model has a fairly low computational cost, demonstrating its efficiency in MI decoding. Table 6 reveals that TriNet has the lowest number of parameters compared to other methods, indicating its compact architecture while achieving comparable or better results. Starting with Shallownet, which has a shallow architecture and the lowest parameter count of 0.047 million, TriNet achieves better classification performance with even fewer parameters. Other methods, such as DeepConvNet and CP-MixedNet, not only have a higher number of parameters compared to TriNet but also exhibit lower EEG decoding accuracy. Overall, TriNet has the lowest parameter count of 0.018 million. Even with this compact architecture, it does not compromise on accuracy. As can be seen from Tables 2, 3, 4, 5 and 6, the outstanding classification performance and low parameter count underscore that TriNet is not only highly robust but also computationally efficient.

C. SUBJECT-INDEPENDENT PERFORMANCE

Table 5 reveals that TriNet has a subject-specific classification accuracy of 92%, whereas the subject-independent classification accuracy is only 78% on the BCI IV-2b dataset. An accuracy loss of nearly 14% is observed in the subject-independent scenario, suggesting that real-time applications of MI classification models still face significant challenges. This is primarily due to inter-subject variations, where different brain regions are activated for the same MI task, leading to reduced accuracy on unseen subject data.

D. APPLICATIONS, LIMITATIONS AND FUTURE DIRECTIONS

The advantages of TriNet are evident from tables 2, 3, 4, 5 and 6, as it provides a comprehensive representation of the EEG signal by extracting features from spectral, spatial, and temporal domains. With its high classification accuracy and low-latency decoding, TriNet shows significant

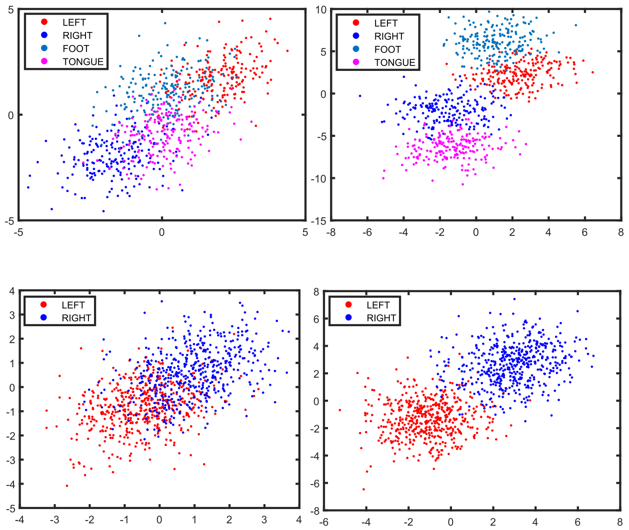

FIGURE 6. Feature discrimination capability of TriNet. The left side shows the raw EEG signal whereas the right side shows the feature separation after TriNet.

potential for several practical BCI applications. For instance, in neuroprosthetics, TriNet can be utilized to ensure precise control of assistive devices to decode user's MI intentions in real-time. Moreover, in applications such as virtual reality or gaming, TriNet could translate the user's motor intentions to enable hands-free control and enhance the user interaction. Additionally, TriNet can assist in neurorehabilitation for patients suffering with a stroke or traumatic brain injury by engaging patients in MI tasks to promote fast recovery.

However, while TriNet has shown commendable performance, it still has some limitations. For instance, despite having high subject-specific classification performance, it struggles to achieve the same levels of classification performance in subject-independent classification as seen in table 5. This suggests that the real-time application of TriNet might face challenges such as reduced performance or signal instability in subject-independent scenarios. Further efforts are needed to make TriNet as robust in subject-independent classification as it is in subject-specific classification. Additionally, while the encoder-decoder architecture has significantly contributed to the classification performance, its convolutional structure introduces several parameters that increase the computational cost. Future work could focus on optimizing the encoder-decoder architecture to further reduce the computational cost and make it more compact while maintaining the same levels of accuracy.

Another critical consideration is the impact of physiological artifacts such as muscle movement, and the user's emotional fluctuations that could significantly impair the MI signal quality while designing real-time BCI applications such as in gaming and neurorehabilitation. Studies such as [46] proposed by Li et al. who introduced a multiscale manifold-based feature fusion method (MMIF) to automatically learn the representation of emotional states, and Babu et al. [47], who proposed the VREEG dataset using immersive reality to enhance the user's emotional states provide promising frameworks to integrate effective emotion recognition into practical EEG systems. In future, TriNet can be extended to operate in parallel with an emotion recognition module, thereby enhancing the robustness and reliability in practical BCI applications. Moreover, TriNet can be adapted to incorporate noise filtering techniques such as multiscale principle component analysis (MSPCA) to further enhance the robustness in real-world applications.

V. CONCLUSION

We proposed TriNet for MI classification by integrating spatial, spectral and temporal features from EEG data. Our approach utilizes FFT to extract spatial features and a spatial transformer to obtain spatial features. Further, an encoder-decoder architecture extracts temporal features which are then concatenated with features from the other two branches to obtain a unified feature representation of the EEG signal. Finally, an ELM classifies the fused feature representation into the respective MI tasks. The highlight of TriNet is its ability to extract features from all three feature domains to obtain a diverse feature representation which contributes to achieving high classification performance. The experimental results signify the efficiency of integrating multi-domain features to extract the complex patterns of MI data with 87.30% and 92.64% accuracy on BCI Competition IV-2a and 2b datasets respectively. Moreover, TriNet also showcases the potential to be utilized for other cognitive tasks and can be extended for several real-life applications. Future work will aim to explore the optimization of TriNet and extend its applications to a wider range of neuroimaging tasks.

DECLARATIONS

ETHICAL APPROVAL

Ethical approval was not sought for the present study because the datasets utilized in this study are publicly available.

DISCLOSURES

The authors declare no potential conflict of interests.

AVAILABILITY OF DATA AND MATERIALS

The dataset underlying the results presented in this paper could be obtained from corresponding author upon reasonable request, and the relevant code of the experiment will be published at https://github.com/

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